The Red Line: How a Map Drawn in 1935 Still Determines Who Builds Wealth in America

One Hundred Years — Report 09 of 10 Window: 1933–2024

Abstract

Between 1935 and 1940, the Home Owners' Loan Corporation graded every neighborhood in 239 American cities on a four-point scale. The criteria were explicit about race. Neighborhoods with Black residents could not receive top grades regardless of housing quality. Grade D — marked in red on the map — meant no federally backed mortgage lending. The federal government then insured $120 billion in mortgages between 1934 and 1962. Two percent went to families of color. This report maps the HOLC grades onto current census data across 224 cities and measures the gap. The median home value in formerly A-graded neighborhoods is $385,000; in formerly D-graded neighborhoods, $185,000. The homeownership gap between Black and white Americans — 29.6 percentage points in 2024 — is wider than it was in 1960 when housing discrimination was still legal. The line was drawn ninety years ago. It is still working.

I. The Grade

In 1933 the federal government created the Home Owners' Loan Corporation to refinance mortgages during the Depression. As a byproduct of that mission, HOLC appraisers walked through 239 American cities and assigned every neighborhood one of four grades:

- **A (Best):** "Homogeneous." "American business and professional men." Green on the map.
- **B (Still Desirable):** "Completely developed." "American families." Blue.
- **C (Declining):** "Infiltration of a lower grade population." Yellow.
- **D (Hazardous):** "Detrimental influences." "Inharmonious racial groups." Red.

The grading criteria are preserved in the HOLC area description forms — standardized government documents that recorded the racial composition, income level, and building quality of every neighborhood surveyed. The forms are explicit. The presence of Black families was itself a negative factor, regardless of income or property condition.

The maps were color-coded. Grade D was red. Redlining.

II. The Amplification

The HOLC maps might have been a historical footnote. They were not, because the Federal Housing Administration adopted their methodology. The FHA's underwriting manual — the document that governed which mortgages the federal government would insure — used the HOLC framework. It directed capital to A and B neighborhoods and away from C and D neighborhoods. It promoted restrictive racial covenants as a positive underwriting factor.

Between 1934 and 1962, the FHA insured $120 billion in mortgages. Two percent went to nonwhite families.

The GI Bill of 1944 compounded the pattern. Out of 67,000 mortgages insured by the GI Bill in New York and northeastern New Jersey in 1950, fewer than 0.8% went to nonwhite veterans. White families bought into Levittown at $7,990 per house. Those houses are now worth approximately $420,000 — a 5,157% return on an investment that was systematically denied to families of color.

The national homeownership rate rose from 45.7% for white families in 1940 to 64.4% by 1960. For Black families it went from 23.6% to 38.4%. The gap widened from 22.1 to 26.0 percentage points during the greatest wealth-building opportunity in American history.

III. The Map Today

The University of Richmond's Mapping Inequality project has digitized and georeferenced the original HOLC maps for all 239 surveyed cities. This report overlays those grades onto current census tract data for 224 cities.

The correlation between a grade assigned in the 1930s and current outcomes is one of the most replicated findings in urban economics:

Metric	Grade A	Grade B	Grade C	Grade D
Median home value	$385,000	$310,000	$225,000	$185,000
Median household income	$82,000	$68,000	$52,000	$41,000
Homeownership rate	71.2%	62.5%	51.8%	41.3%
Life expectancy	80.4 yrs	78.9 yrs	77.1 yrs	75.2 yrs
College attainment	48.5%	38.1%	27.6%	19.2%
Poverty rate	8.2%	12.4%	18.9%	27.3%
Percent minority	22.8%	33.4%	48.7%	63.1%
Tree canopy coverage	38.2%	31.6%	24.3%	18.4%

The gradient is monotonic across every dimension. Neighborhoods graded A in 1935 have double the home values, half the poverty rate, and five additional years of life expectancy compared to neighborhoods graded D. The line drawn on a government map ninety years ago predicts who lives longer today.

IV. The Wealth Gap

The homeownership gap is the mechanism through which the redlining era continues to compound. Homeownership is the primary wealth-building instrument for most American families. Families excluded from homeownership in the postwar era did not just miss a house purchase — they missed every dollar of appreciation that house would have generated across three generations.

The racial wealth gap data tells the story:

Year	White median wealth	Black median wealth	Ratio
1963	$56,000	$6,000	9.3:1
1989	$112,000	$9,000	12.4:1
2010	$113,000	$6,000	18.8:1
2022	$285,000	$44,900	6.3:1

The 2010 figure is the most revealing. The subprime mortgage crisis — which disproportionately targeted Black homeowners for predatory lending products — erased a generation of Black wealth accumulation in three years. White median wealth fell 26%; Black median wealth fell 65%. The ratio hit 18.8:1, the worst on record.

By 2022 the ratio had narrowed to 6.3:1, driven primarily by appreciation in housing values. But the homeownership gap itself — the structural mechanism — remains wider than it was when housing discrimination was legal:

- 1940: 22.1 percentage points (white 45.7%, Black 23.6%)
- 1960: 26.0 points (64.4% vs 38.4%)
- 2000: 26.6 points (73.8% vs 47.2%)
- 2024: 29.6 points (74.5% vs 44.9%)

The gap has widened by 7.5 percentage points since 1940. The Fair Housing Act of 1968 made discrimination illegal. The structural consequences of the discrimination that preceded it continue to widen.

V. The Tree Canopy

One of the most striking correlations in the data is not economic. It is ecological. Neighborhoods graded A in the 1930s have 38.2% tree canopy coverage today. Neighborhoods graded D have 18.4%. The line drawn around a neighborhood ninety years ago predicts how many trees it has.

This is not coincidental. HOLC A neighborhoods received sustained public and private investment — parks, street trees, infrastructure maintenance. D neighborhoods received disinvestment. The physical infrastructure follows the capital. Trees follow investment. Shade follows trees. Surface temperature follows shade. Health outcomes follow temperature.

The tree canopy disparity is a measure of the cumulative environmental consequence of ninety years of differential investment, visible from space and measurable in degrees of summer surface temperature.

VI. Limitations

1. **Selection effects.** HOLC did not randomly assign grades. The neighborhoods that received D grades had pre-existing disadvantages that were correlated with but not caused by the grade itself. Separating the causal effect of the HOLC grade from pre-existing conditions requires careful identification strategy.

2. **Geographic change.** Census tract boundaries have changed multiple times since the 1930s. Crosswalking HOLC boundaries to current census geography introduces

measurement error. We use area-weighted interpolation following the methodology established by the Mapping Inequality project.

3. **The counterfactual.** We cannot observe what these neighborhoods would look like without redlining. The policy was applied to every major American city simultaneously. There is no untreated control group.

4. **Data coverage.** HOLC surveyed 239 cities but not all American cities. The cities surveyed were disproportionately large and in the Northeast and Midwest. The findings may not generalize to cities that were not surveyed.

5. **Causal complexity.** Redlining was not the only mechanism of housing discrimination. Restrictive covenants, steering, blockbusting, predatory lending, exclusionary zoning, and individual discrimination all contributed to the patterns documented here. Attributing current outcomes to HOLC grades specifically overstates the role of a single policy instrument.

VII. Conclusion

The HOLC maps were drawn as an administrative tool for mortgage lending. They became the template for the largest government-directed wealth transfer in American history — $120 billion in federally insured mortgages directed overwhelmingly to white families and white neighborhoods.

The Fair Housing Act made the practice illegal in 1968. By then, white homeownership rates had reached 68% while Black homeownership remained at 44%. The wealth generated by three decades of subsidized homeownership — the appreciation, the equity, the intergenerational transfers — went to families on one side of the line. Families on the other side watched the same houses appreciate from neighborhoods where they could not buy.

The data does not argue for any particular remedy. It documents a specific fact: a line drawn on a government map in 1935 predicts who has wealth, who owns a home, who lives longer, and who has trees on their street in 2024. Ninety years. The gradient is monotonic. The line is still working.

Acknowledgments

Data sourced from the University of Richmond's Mapping Inequality Project (HOLC maps and area descriptions), the U.S. Census Bureau (American Community Survey), the Federal Reserve's Survey of Consumer Finances (wealth data), the Bureau of Labor Statistics, and the National Center for Health Statistics. HOLC area descriptions digitized by the Digital Scholarship Lab at the University of Richmond under CC BY-NC-SA.

Any errors of methodology, interpretation, or framing are the author's alone.

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